

Experiment:-8 Nmap Service Version and OS Detection in Kali Linux

Aim

To execute Nmap commands in Kali Linux to detect service versions, perform an aggressive scan, and identify the operating system of a target machine.

Procedure

1. Open the Kali Linux terminal.
2. Ensure Nmap is installed using `nmap --version`.
3. Select an authorized target IP address, such as `192.168.1.10`.
4. Execute the commands given below one by one.
5. Observe the scan results, including open ports, service versions, and operating system details.

Commands / Code

i) Detect the version of services running

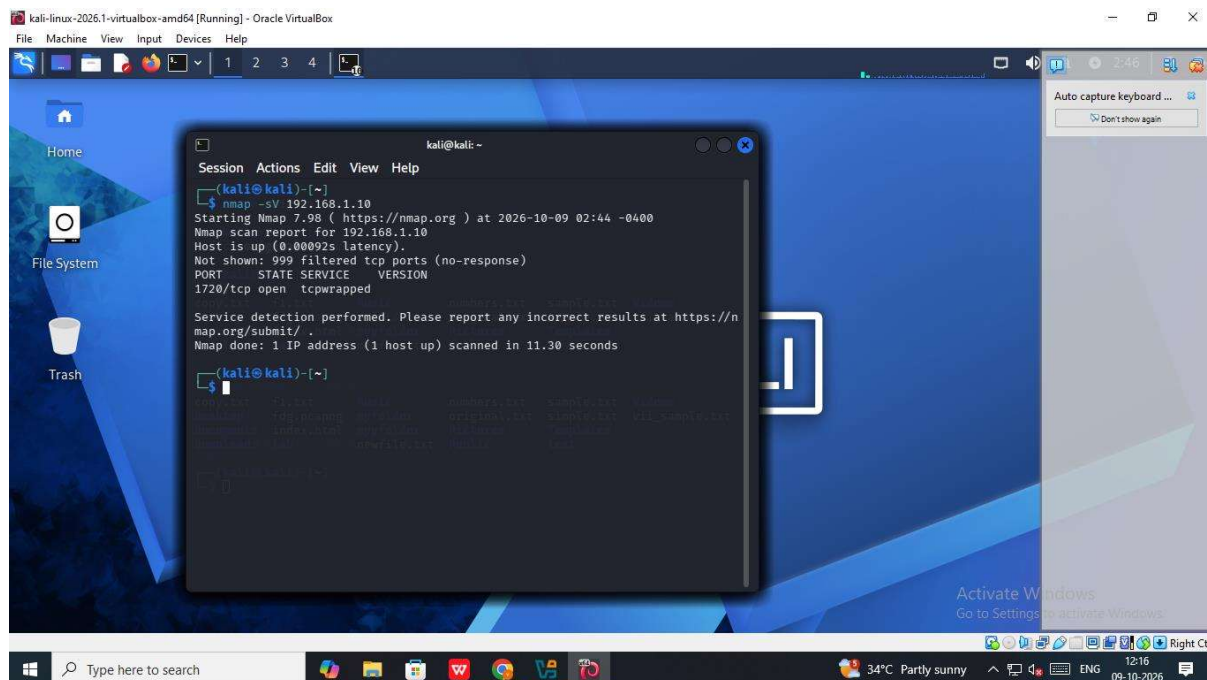
Command:

```
nmap -sV 192.168.1.10
```

Sample Input:

```
nmap -sV 192.168.1.10
```

Sample Output (illustrative):



ii) Aggressive scan

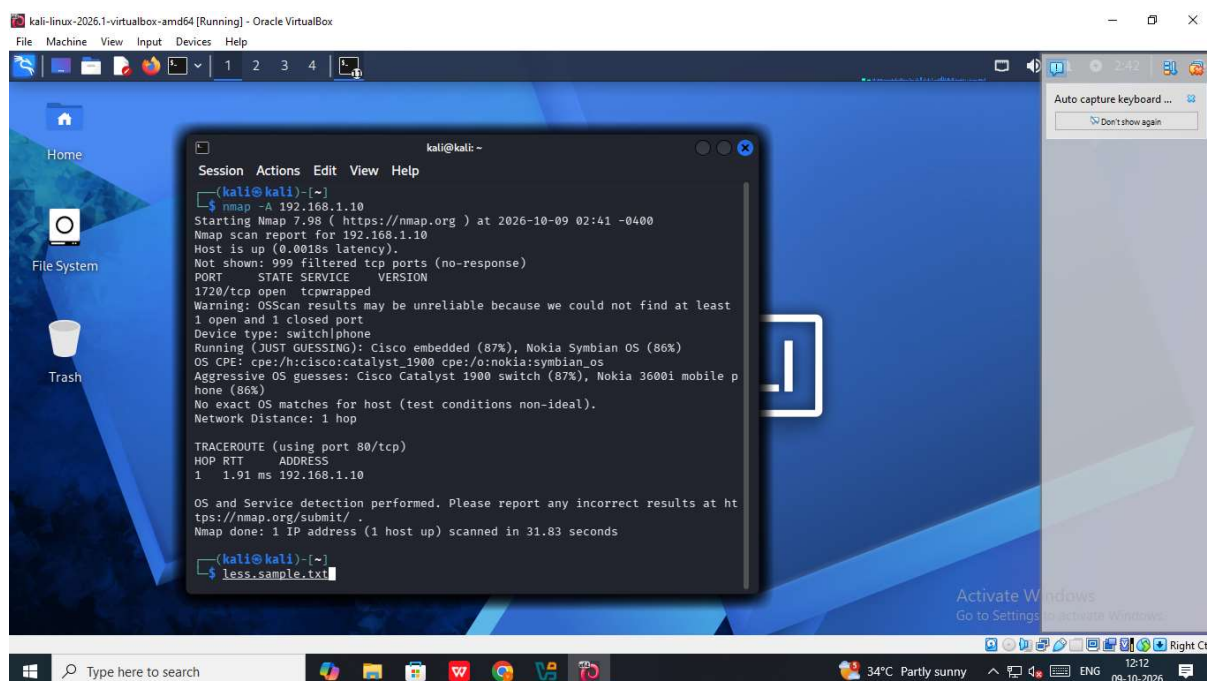
Command:

```
nmap -A 192.168.1.10
```

Sample Input:

```
nmap -A 192.168.1.10
```

Sample Output (illustrative):



iii) Detect the operating system of the target

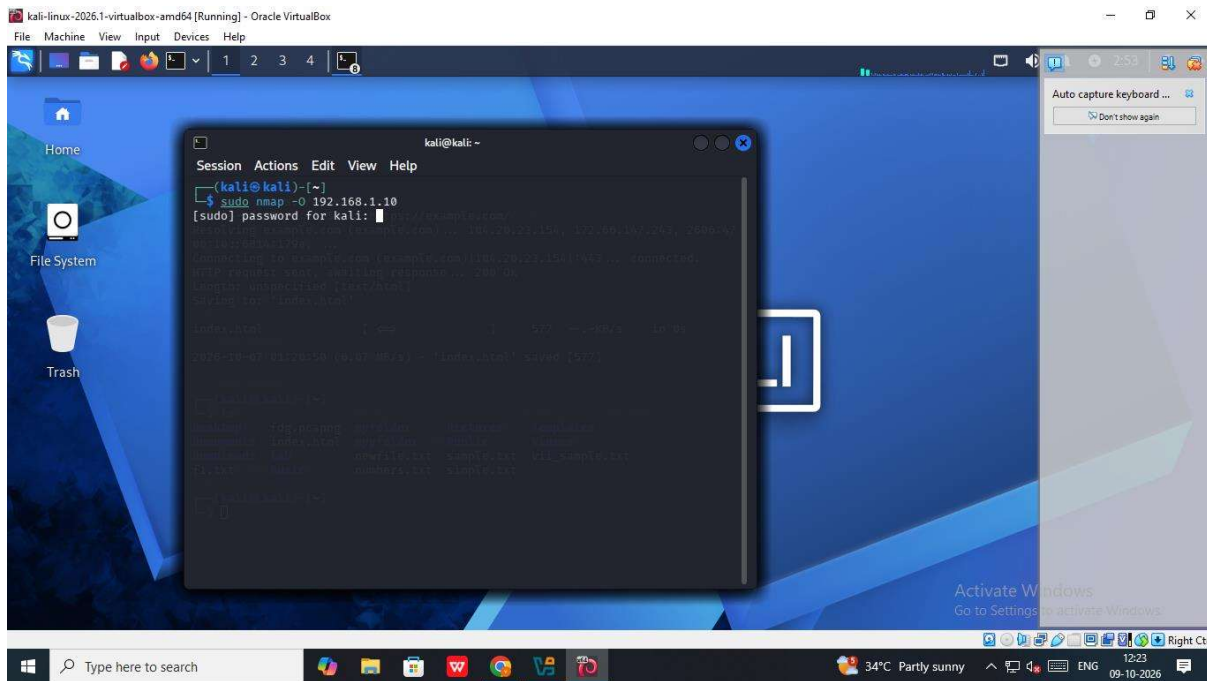
Command:

```
sudo nmap -0 192.168.1.10
```

Sample Input:

```
sudo nmap -O 192.168.1.10
```

Sample Output (illustrative):



Result

Nmap commands were used to demonstrate how to:

1. Detect running service versions using `nmap -sV`.
2. Perform an aggressive scan using `nmap -A`.
3. Identify the target operating system using `sudo nmap -O`.

Conclusion: Nmap helps identify network services, software versions, and probable operating systems for authorized security testing.